Machine Learning Pipelines

AIML231 Week 4

Dr Emily O'Riordan

emily.o'riordan@vuw.ac.nz

Today’s Lecture

ML Pipeline Motivation

What is an ML Pipeline?

CRISP-DM

Modern ML Pipelines & MLOps

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ML Pipelines Motivation

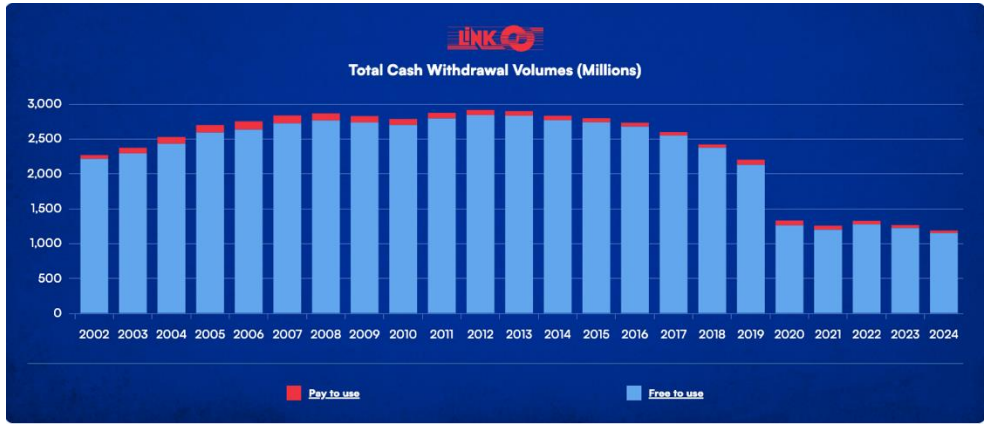
You are an expert ML engineer and you are hired to build ML models for two companies.

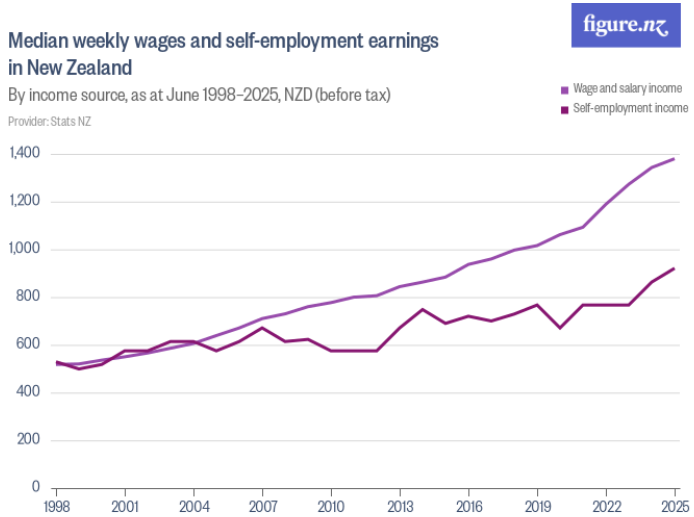
- Company A's model *classifies flower species from images*
- Company B's model *detects fraudulent bank transactions*

Will you trust these models in one year?

Model Staleness

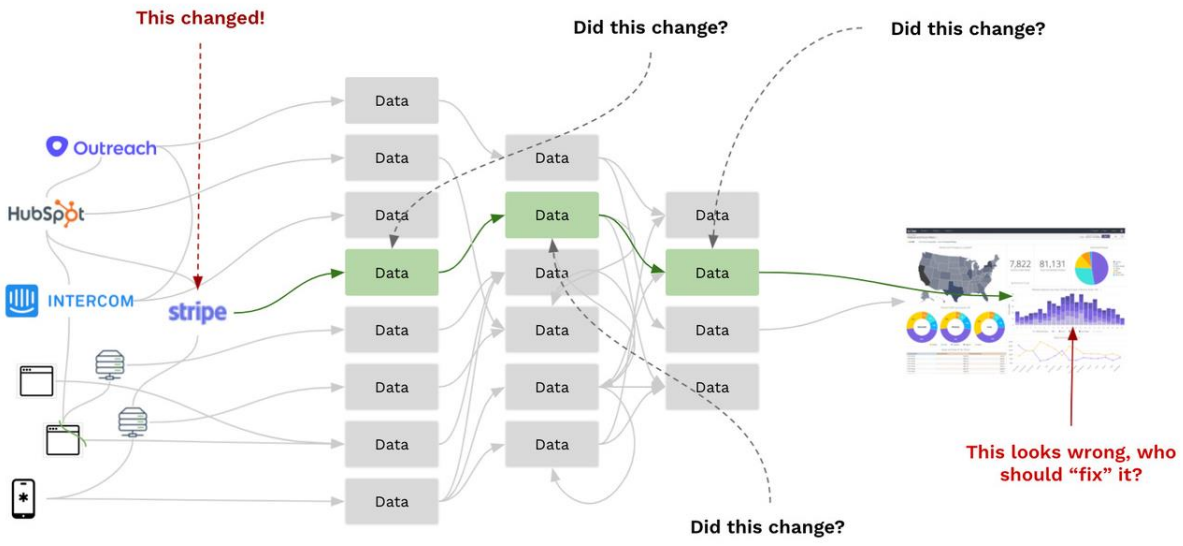
- ML models can become outdated due to changes in data distributions
- Model staleness: when a deployed model no longer reflects the current system it operates in
- Data drift: when the input data distribution $P(x)$ shifts
  - More online or contactless payments
  - Transaction amounts increase over time
- Concept drift: when the relationship between the input data and the target data $P(y|x)$ changes
  - New fraud strategies emerge
  - Salary categories (low, medium, high) change due to inflation





“Let’s just retrain the model!”

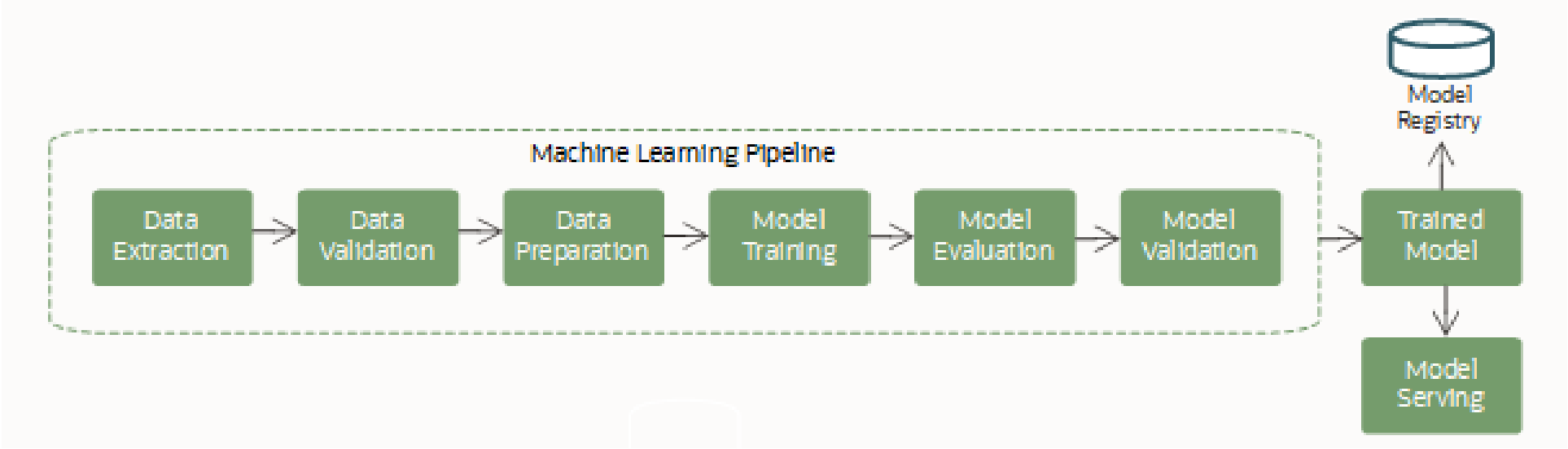
- Not as easy as that…
- Model retraining has many steps and challenges:
  - Data comes from multiple sources, with changing formats, schemas, and quality
  - Preprocessing steps (cleaning, feature extraction) must be **identical** every time
  - Software and environments change (libraries, Python versions, hardware)
  - Models may behave differently across systems (e.g., GPU vs CPU, precision differences)



Why we need ML pipelines

If we **standardise** and **automate** these processes, we can ensure:

- Repeatability
- Observability
- Modularisation
- Efficiency
- Safety



Today’s Lecture

ML Pipeline Motivation ✅

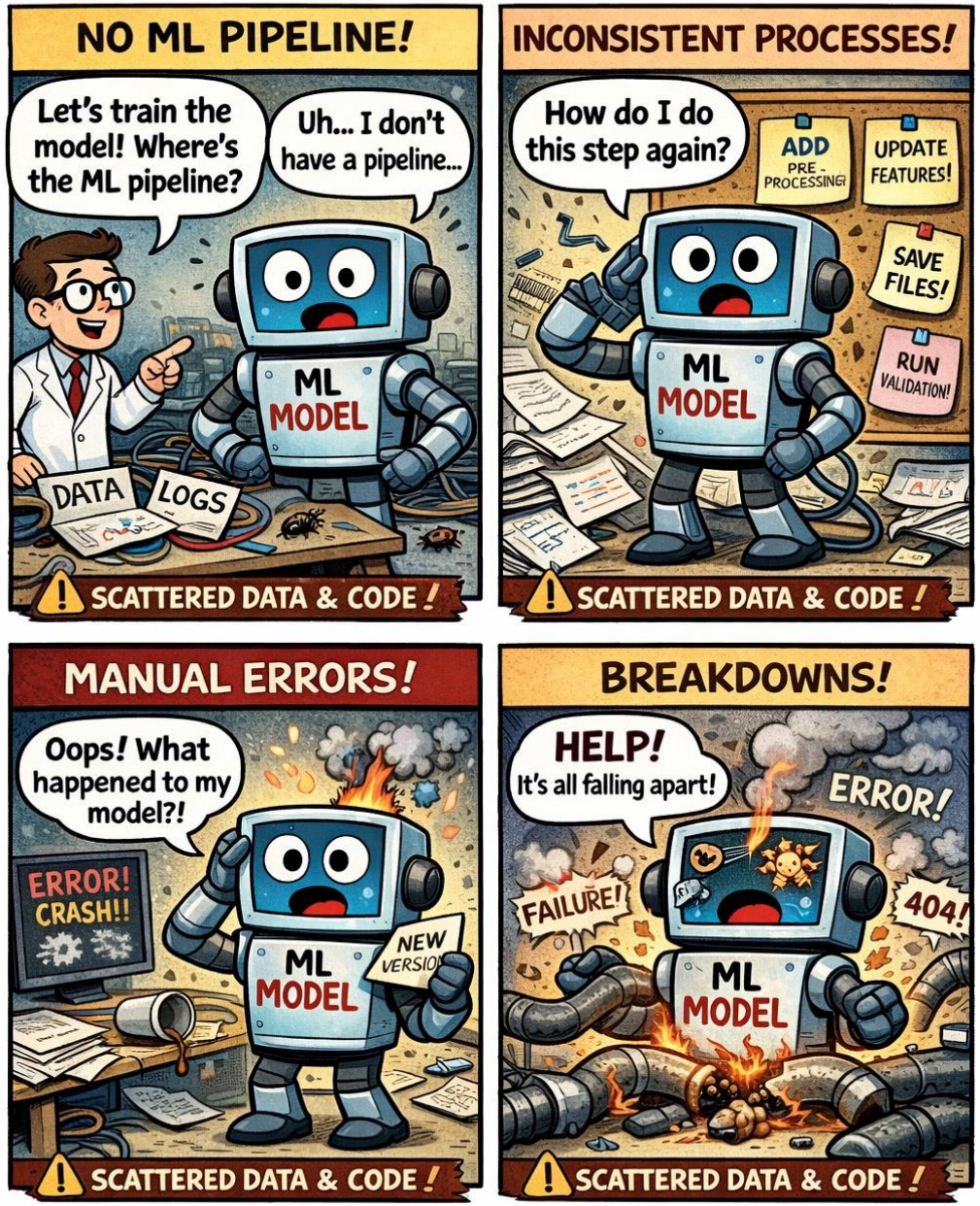
What is an ML Pipeline?

CRISP-DM

Modern ML Pipelines & MLOps

What is an ML Pipeline?

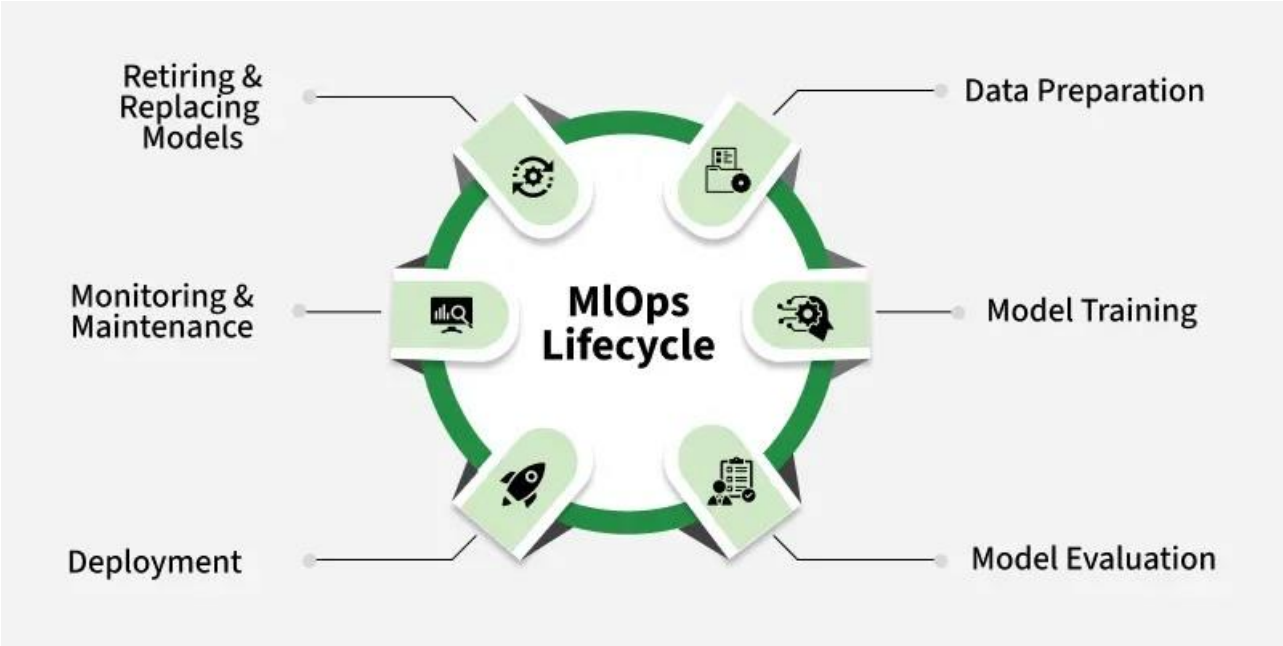
- ML in the real world is about more than just the model!
- An ML pipeline is everything required to make an ML model work *reliably* in the real world, including steps before, during, and after training.
- A sequence of automated, repeatable steps that streamline the model lifecycle.



What is an ML Pipeline?

A sequence of automated, repeatable steps that streamline the model lifecycle.

- Data collection, ingestion, validation
- Data preprocessing and feature selection
- Model training and evaluation
- Deployment and monitoring
- Feedback for retraining or rollback



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ML Pipeline Motivation ✅

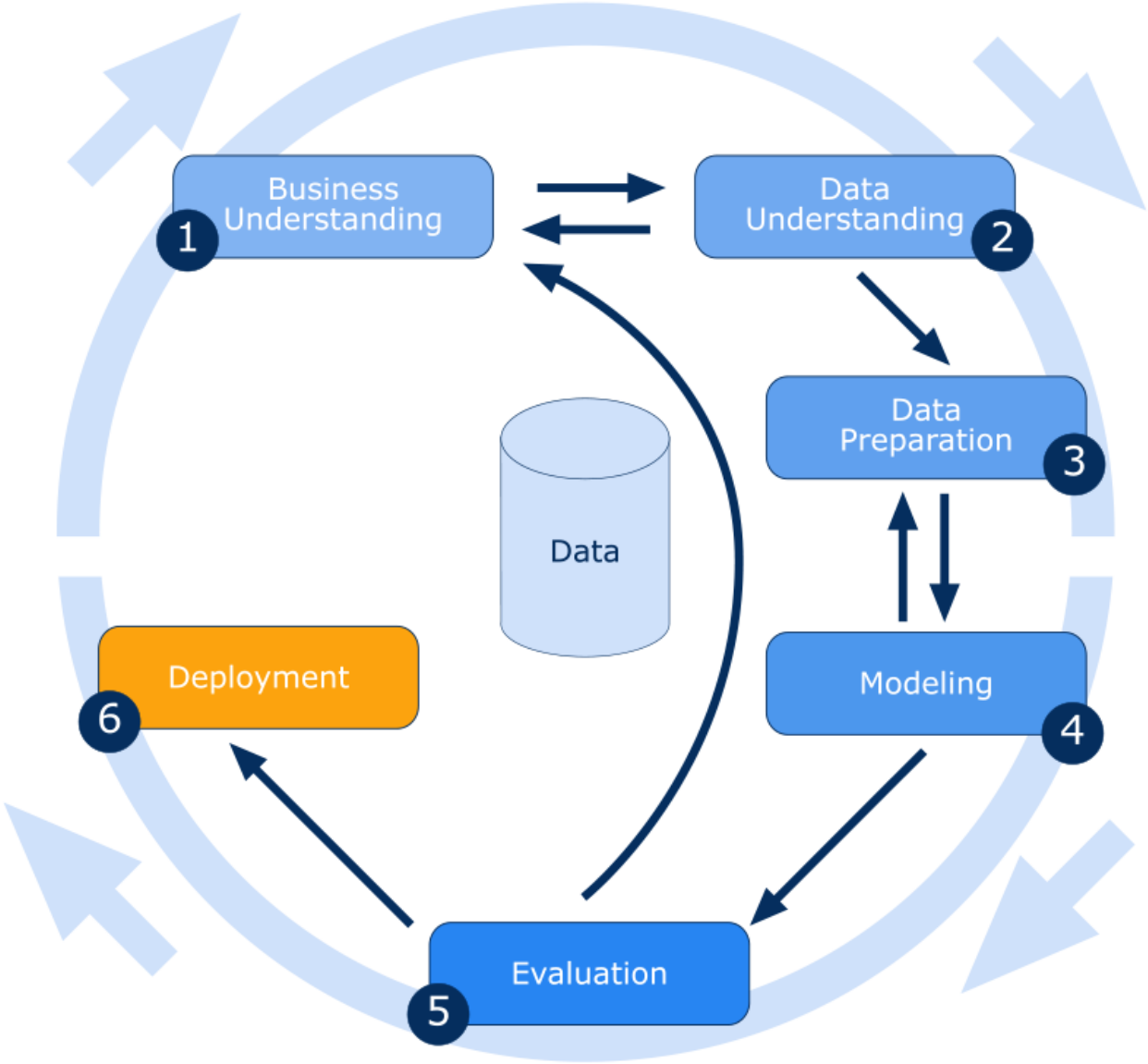
What is an ML Pipeline? ✅

CRISP-DM

Modern ML Pipelines & MLOps

CRISP-DM

- **CR**oss **I**ndustry **S**tandard **P**rocess for **D**ata **M**ining (CRISP-DM)
- Developed by a consortium of data analysis companies *in 1996*
- Still a standard methodology for *data science* projects

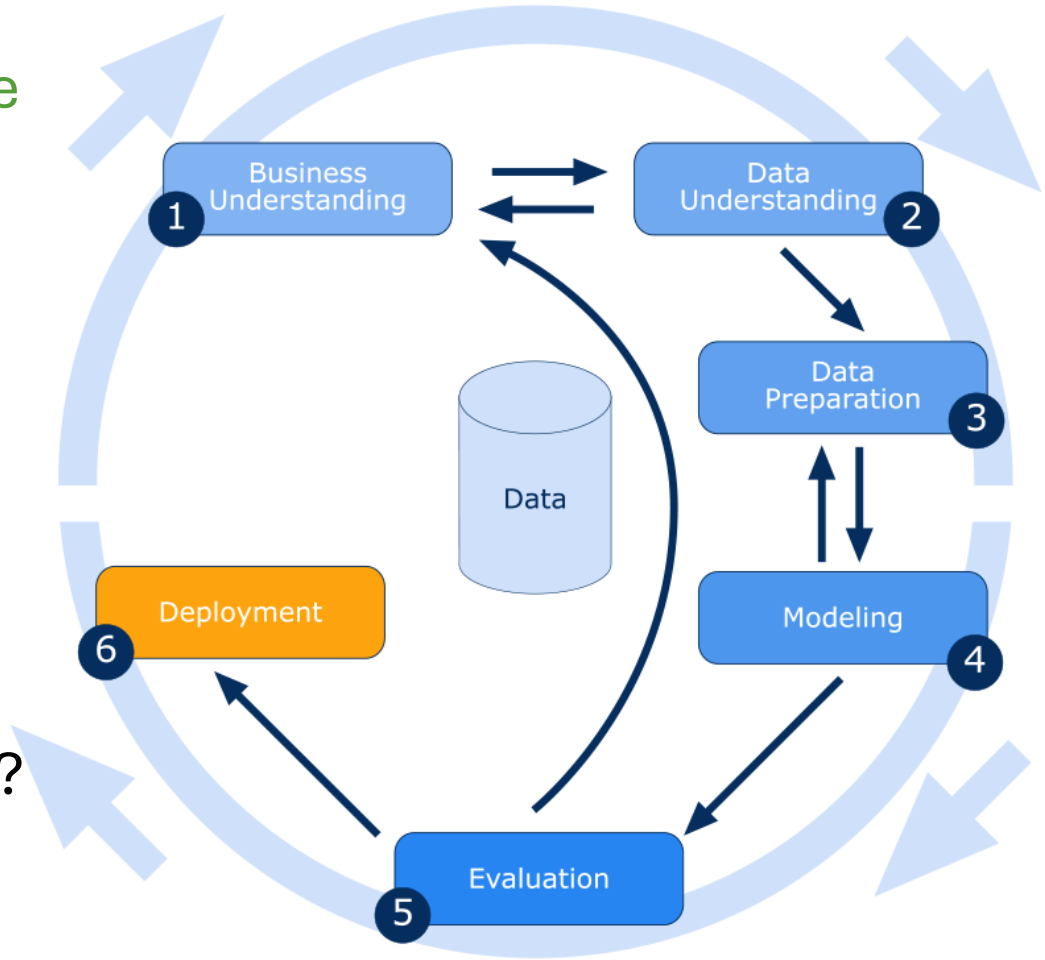


CRISP-DM: Business Understanding

Working with the customer/end user to determine the objectives and requirements of the project.

Made up of four tasks:

- Determine business objectives: What is the success criteria?
- Assess current situation: What resources are available? Any risks, contingencies, ...
- Determine ML success criteria: What metrics? What does success look like in technical terms?
- Produce project plan: What tools? How are you going to do it? Gantt charts?

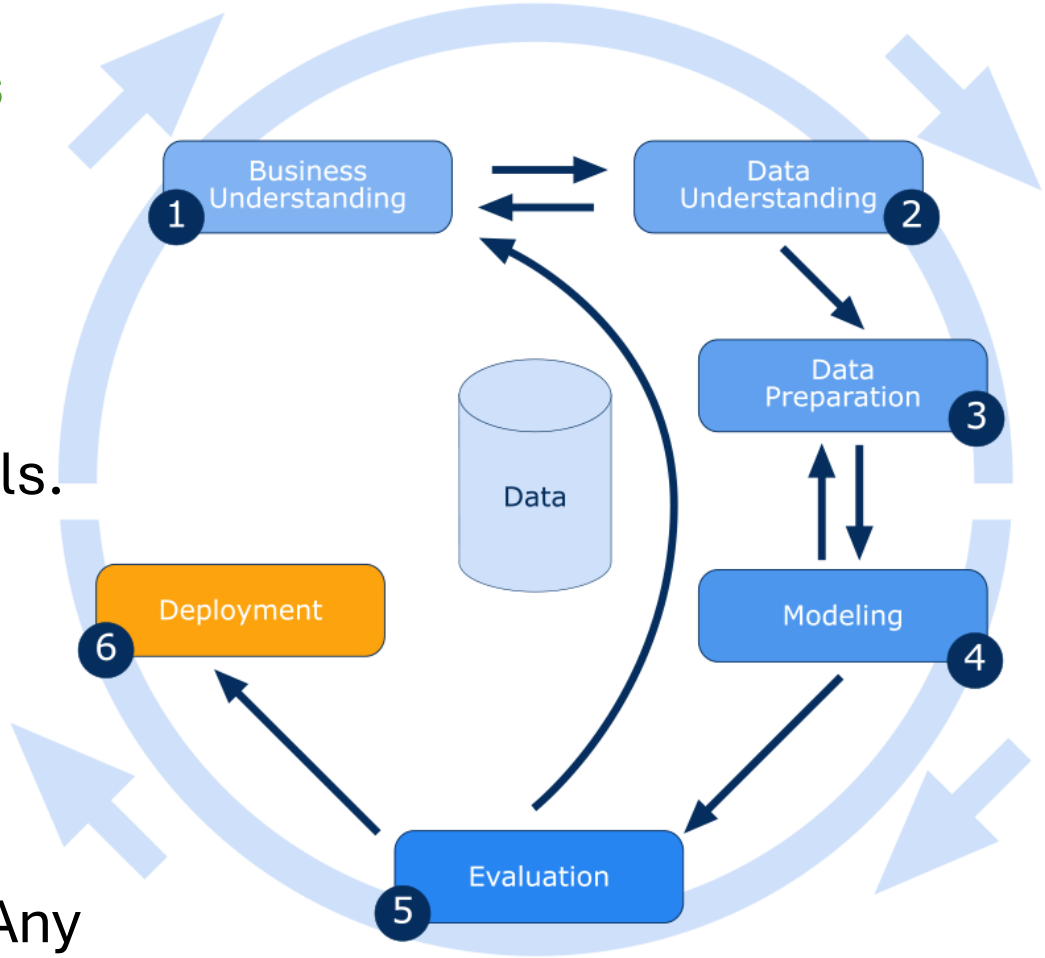


CRISP-DM: Data Understanding

Identifying, collecting, and analysing the data sets that can help you accomplish the project goals.

Made up of four tasks:

- Collect initial data: Acquire the necessary data and write pipelines to load into Python/other tools. There may be multiple sources to integrate.
- Describe data: Examine and document the metadata, e.g., format, number of records, ...
- Explore data: Dig deeper into the data. Query it, visualise it, and identify relationships.
- Verify data quality: How clean/dirty is the data? Any missing data? Document any quality issues.

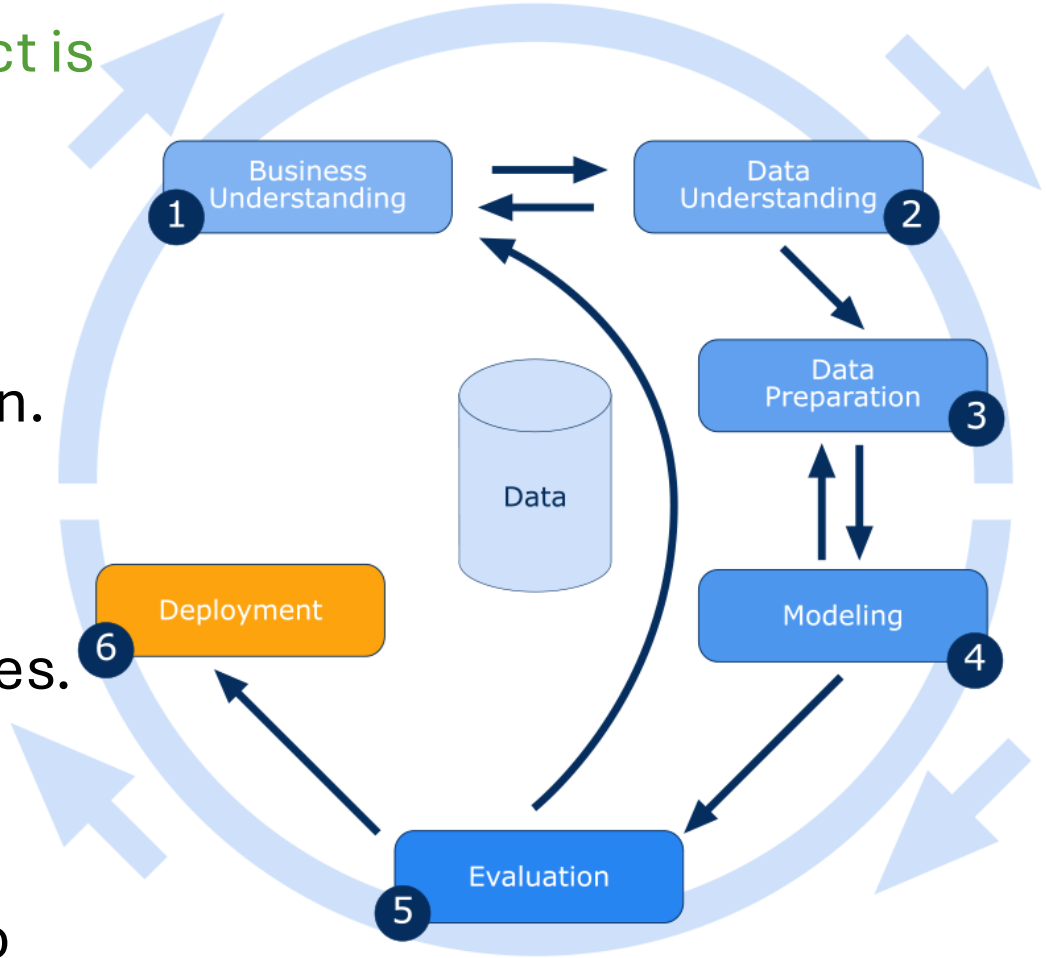


CRISP-DM: Data Preparation

A common rule of thumb is that 80% of the project is data preparation. Tomorrow's lecture!

Made up of five tasks:

- Select data: Determine which data sets will be used, document reasons for inclusion/exclusion.
- Clean data: Prevent garbage-in, garbage-out! Correct, impute, or remove erroneous values.
- Construct data: Feature engineering, e.g., normalising, transforms, or creating new features.
- Integrate data: Combine data from multiple sources.
- Format data: Ensure data is in the right format, e.g., convert string values that store numbers to floats.



CRISP-DM: Modelling

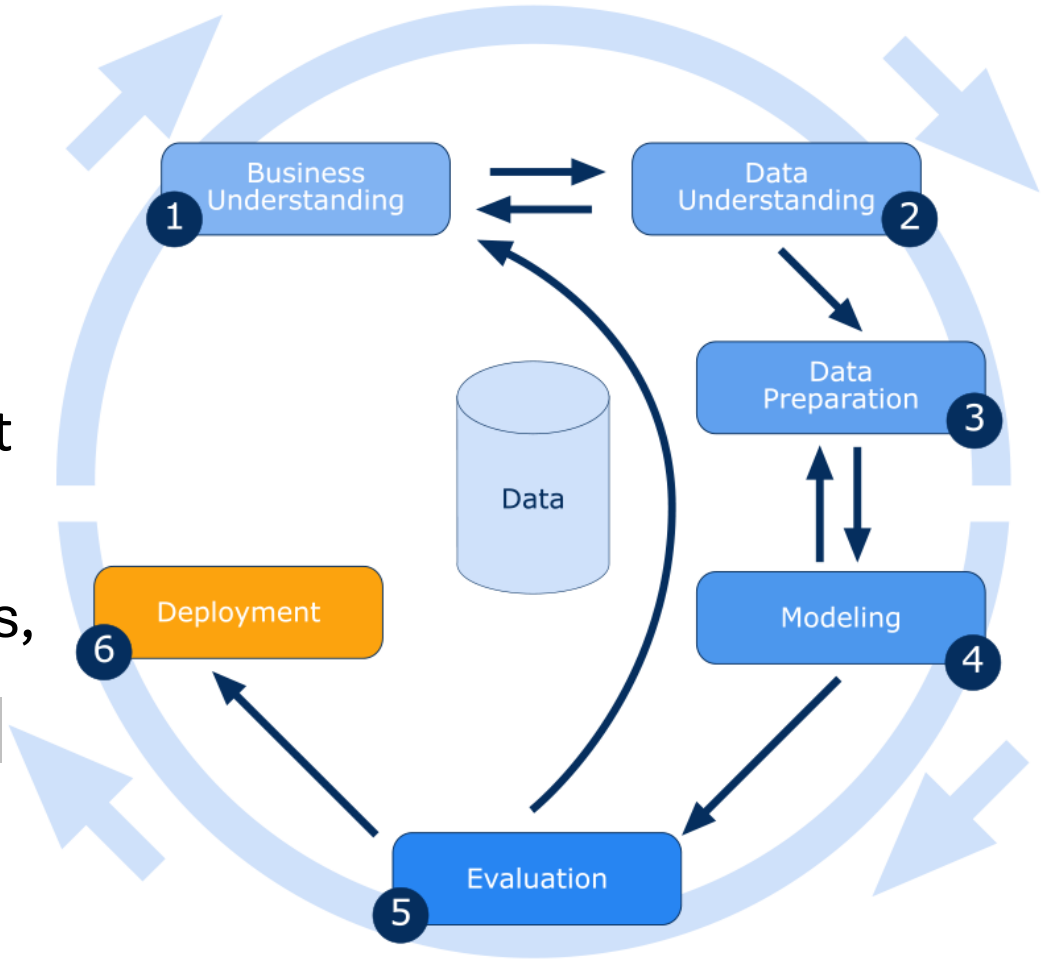
Build and assess various ML models.

Made up of four tasks:

- Select model architecture(s): Determine which algorithms to try (e.g., regression, neural net).
- Generate test design: Generate a procedure to test the model's quality (e.g., separate data into training and test).
- Build model: Choose (and tune) hyperparameters, run the code!

```
model = LinearRegression().fit(X_train, y_train)
```

- Assess model: Interpret the model results based on domain knowledge, the pre-defined success criteria, and the test design.

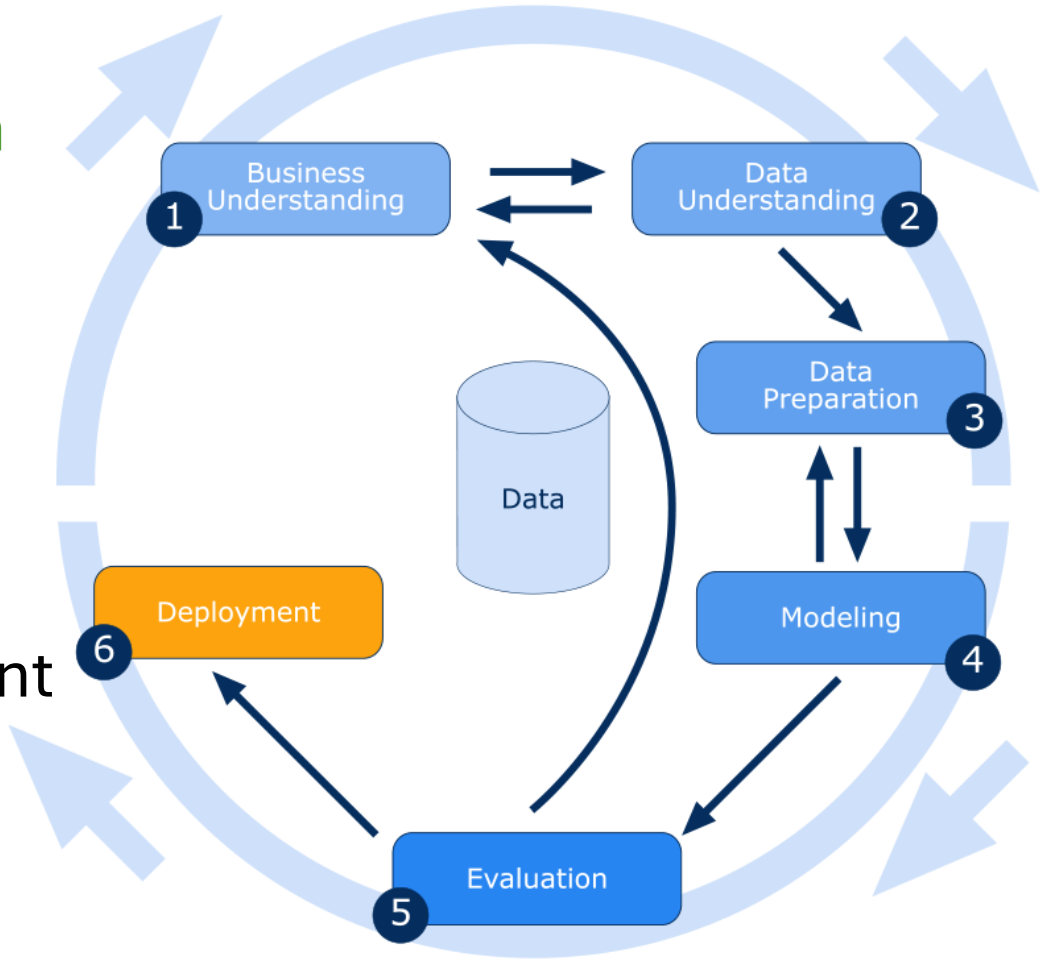


CRISP-DM: Evaluation

Whereas the the previous phase focuses on technical evaluation, this phase looks at which model best meets the business needs.

Made up of three tasks:

- Evaluate results: Do the models meet the business success criteria? Which one(s) should we approve for the business?
- Review process: Review the work accomplished. Summarize findings, document methods, and correct anything if needed.
- Determine next steps: Based on the previous three tasks, determine whether to proceed to deployment or iterate further.

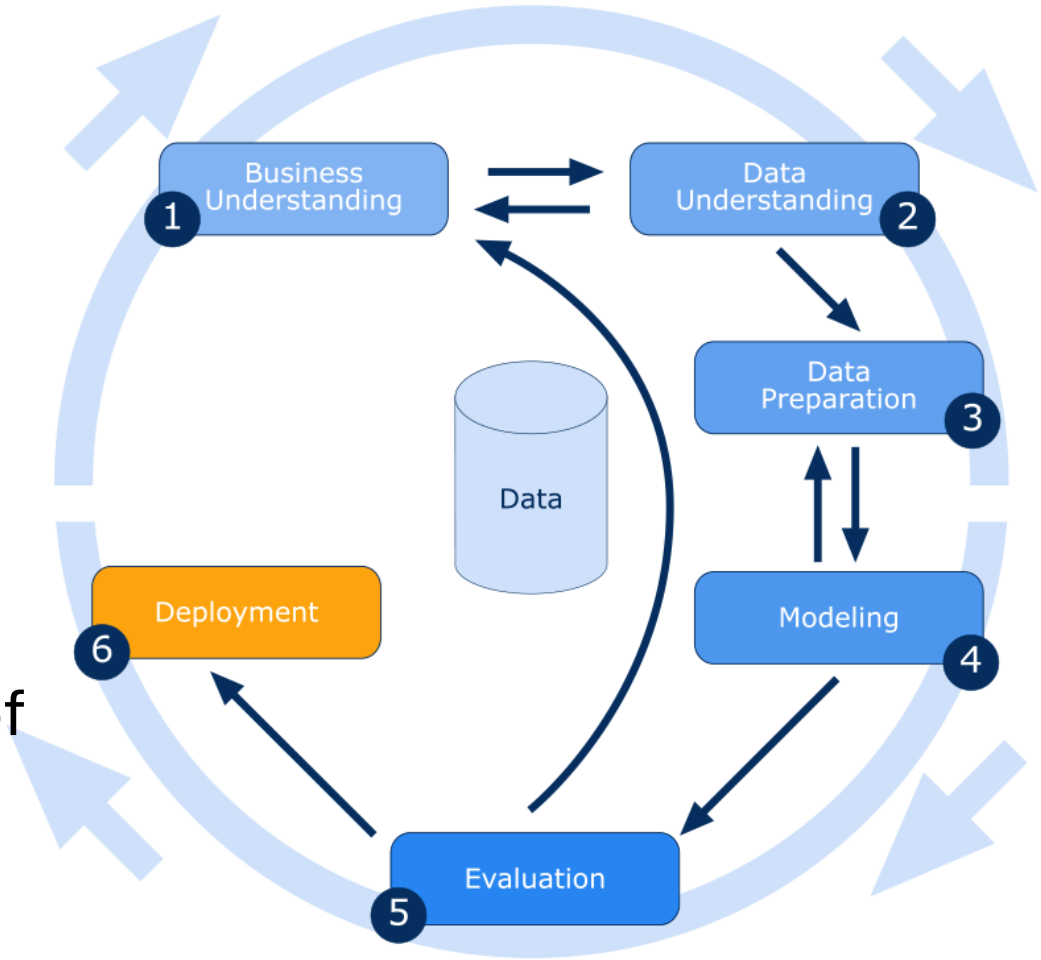


CRISP-DM: Deployment

Making the model operational and its results accessible by the end user.

Made up of four tasks:

- Plan deployment: Develop and document a plan for deploying the model.
- Plan monitoring and maintenance: Develop a thorough plan to avoid issues during operation.
- Produce final report: Document a summary of the project, including ML/data mining results.
- Review project: Conduct a review of what went well, what could have been better, and how to improve in the future.



Today’s Lecture

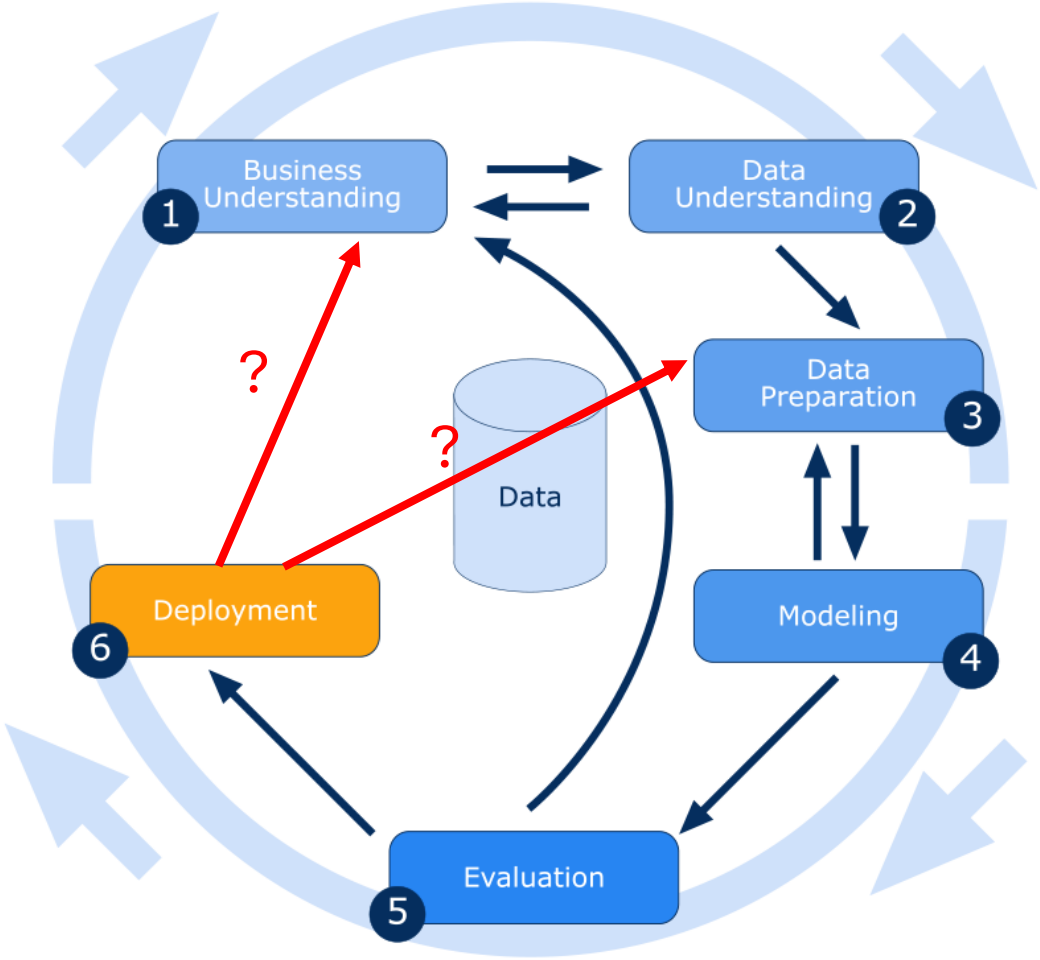
ML Pipeline Motivation ✅

What is an ML Pipeline? ✅

CRISP-DM ✅

Modern ML Pipelines & MLOps

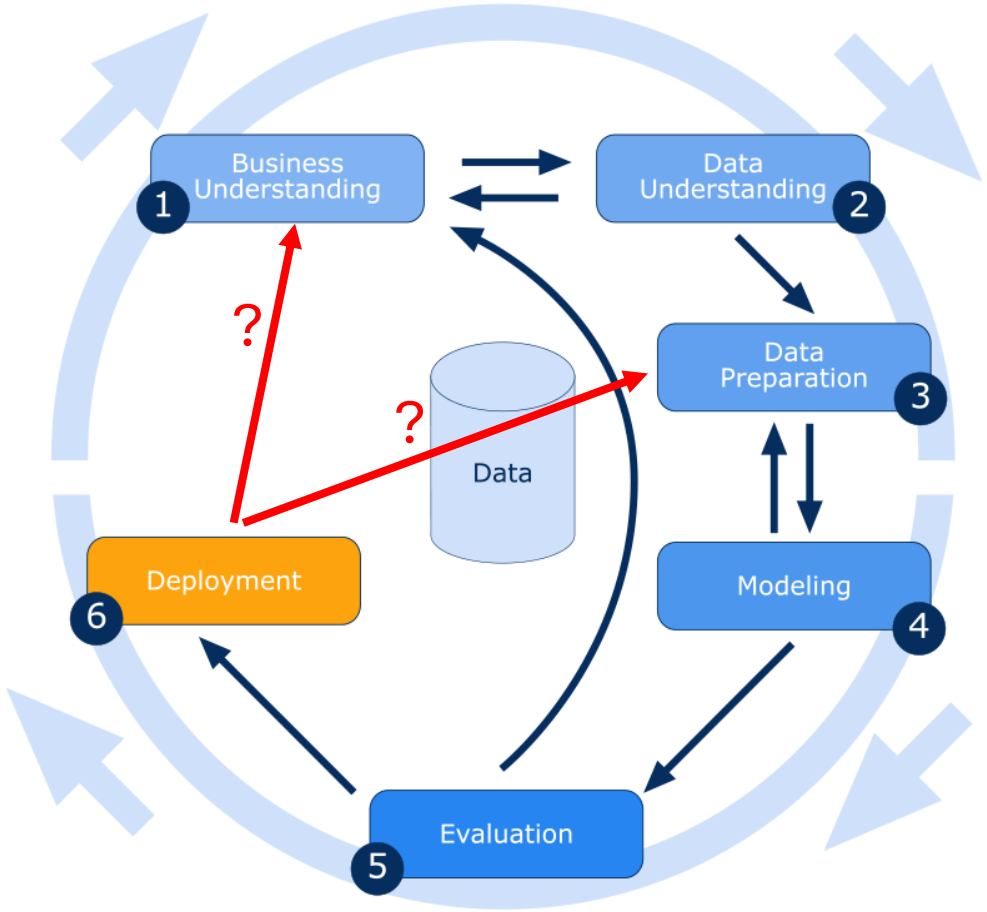
What’s missing for ML projects?



What happens after deployment!

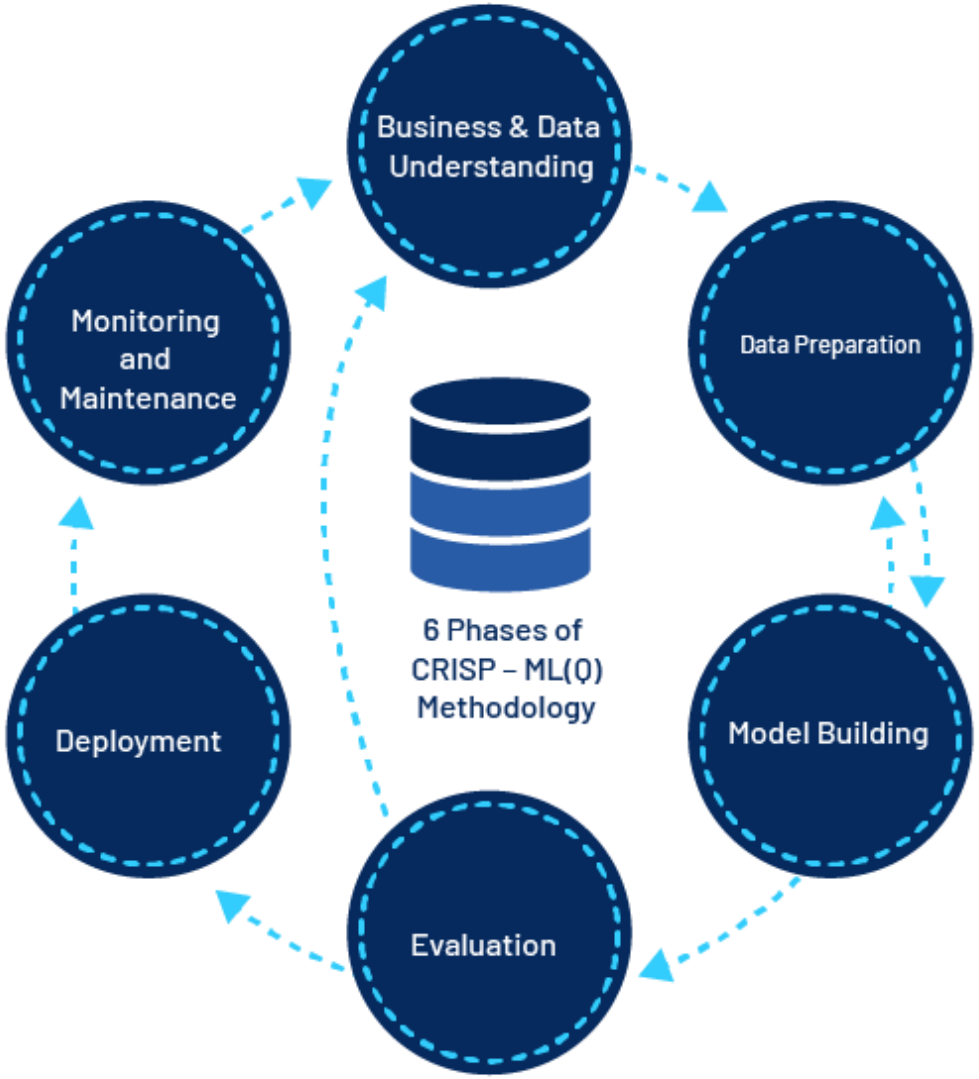
Modern MLOps

Data Science Projects	Machine Learning Projects
Deployment of the model is the end of the project	Deployment of the model rarely the end of the project
Model is static	Data and concept drift
One-off project with set data	Continuous system with updating data



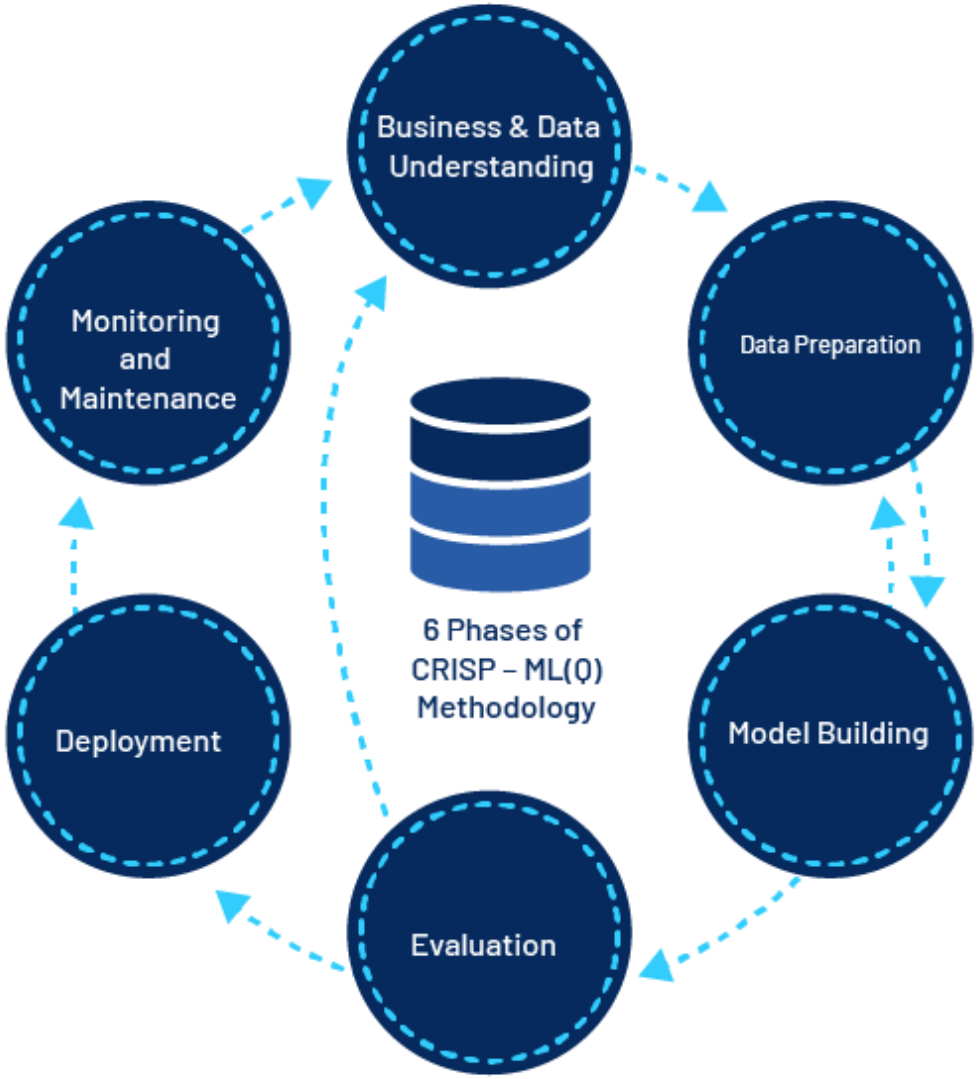
Modern MLOps: Monitoring and Maintenance

- A monitoring phase should be added to the pipeline to track metrics (e.g., accuracy, error, data distributions) in operation to detect data/concept drift in production.
- This should log outcomes, alert when there are anomalies, and evaluate model quality over time.
- If monitoring shows degradation, the team can decide to retrain, revert, or update the model.

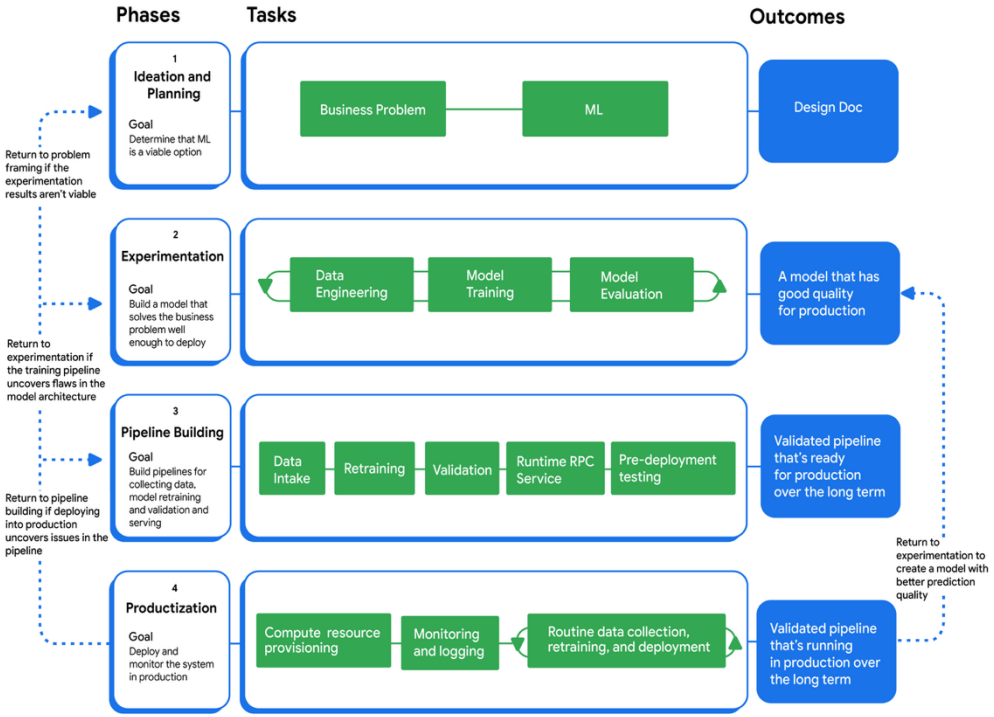


Modern MLOps: Continuous Retraining (optional)

- When new data becomes available or when monitoring signals that performance is falling below a threshold, the system could trigger a retraining of the model with up-to-date data.
- Data ingestion and preparation steps are automated, so that new data can regularly feed into the pipeline.
- Today, companies aim to "recycle" the pipeline automatically: for example, re-run training monthly on the latest data, or retrain whenever a significant drift in data is detected.
- Automated pipelines as much as possible – less manual errors, more consistent methodology, faster systems.

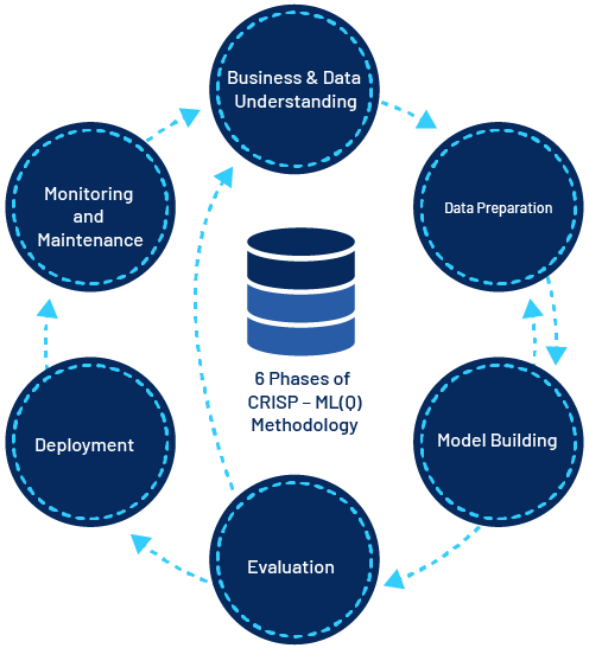


Modern ML Pipeline

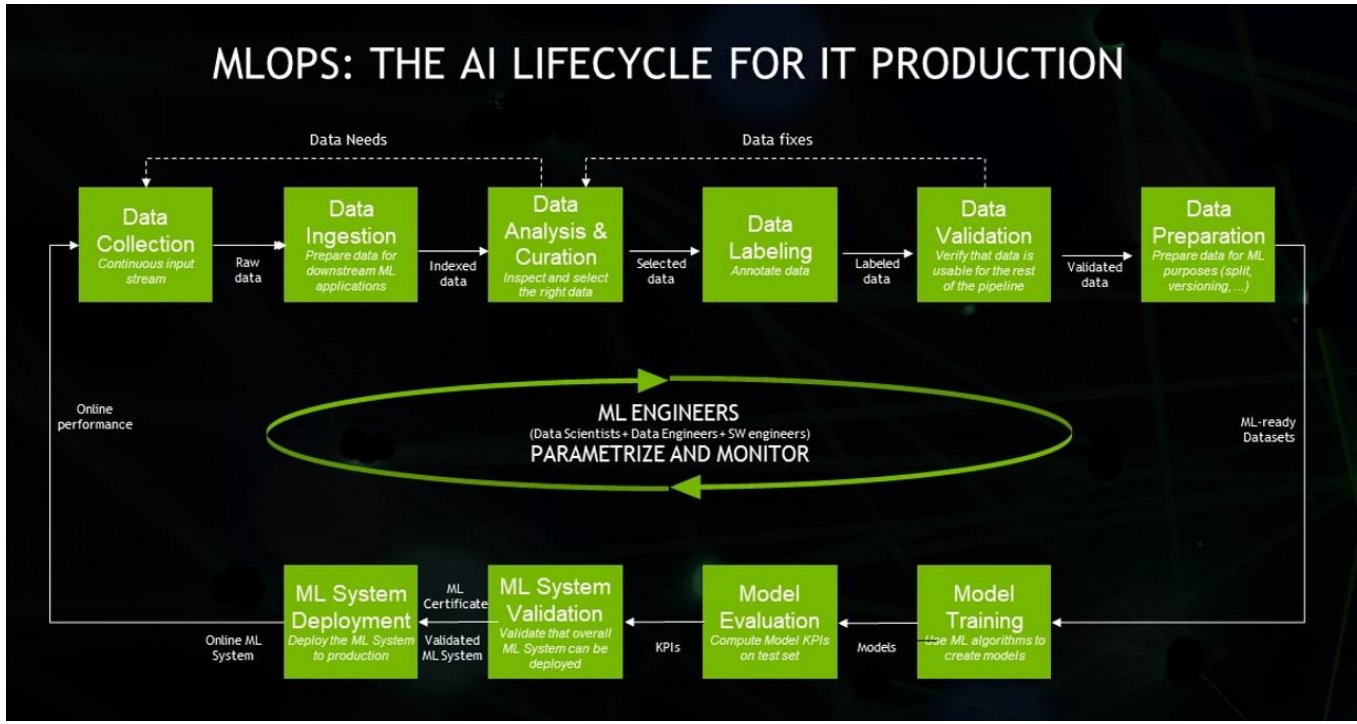


Google ML dev

CRISP-ML(Q)



NVIDIA MLOps



Research vs Production Pipelines

	Research	Production
Team	Small number of researchers	ML engineers, software developers, project manages
Goal	Exploration, experimentation, understanding	Reliable, repeatable delivery
Consequences of a broken pipeline	Annoying, slows progress	Business interruption
Reproducibility	Important for papers and assignments	Important for safety, auditing, rollback
Scale	Small-to-moderate datasets, few users	Large datasets, many users

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- ML Pipeline Motivation ✅
- What is an ML Pipeline? ✅
- CRISP-DM ✅
- Modern ML Pipelines & MLOps ✅

Further Reading

- NVIDIA MLOps
- Google ML dev
- CRISP-ML(Q)
- MLOps vs DevOps
- CRISP-DM